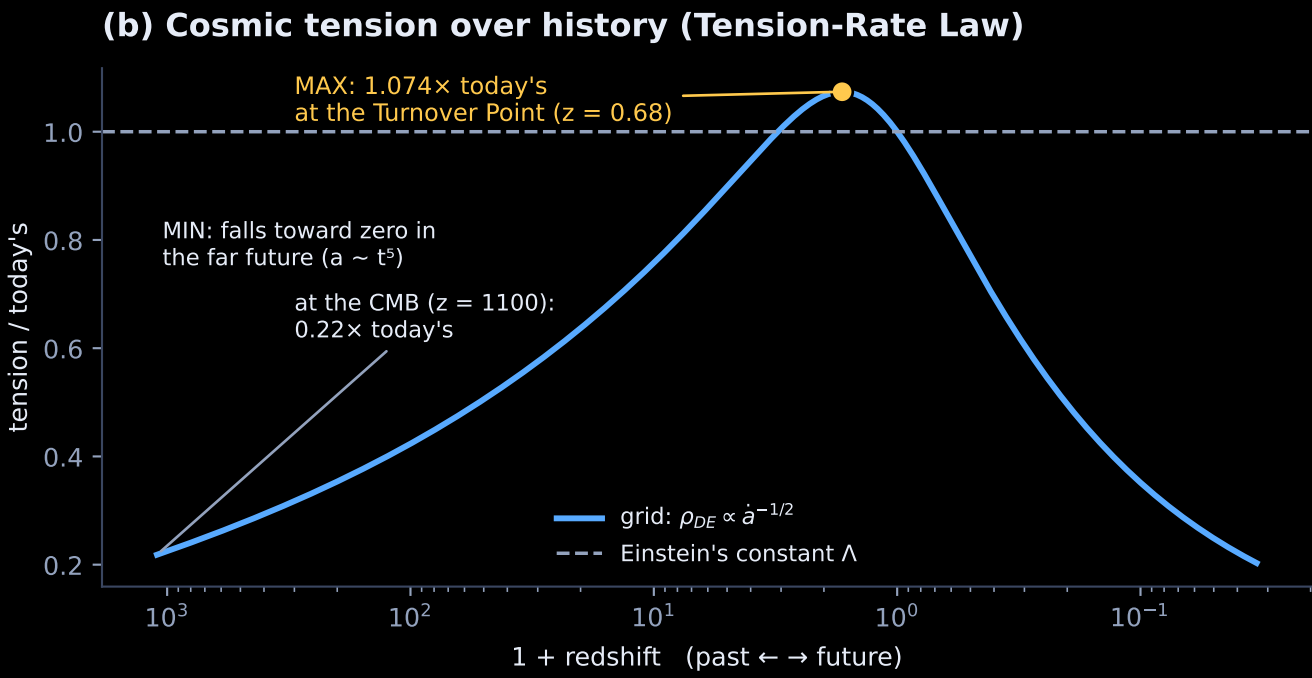
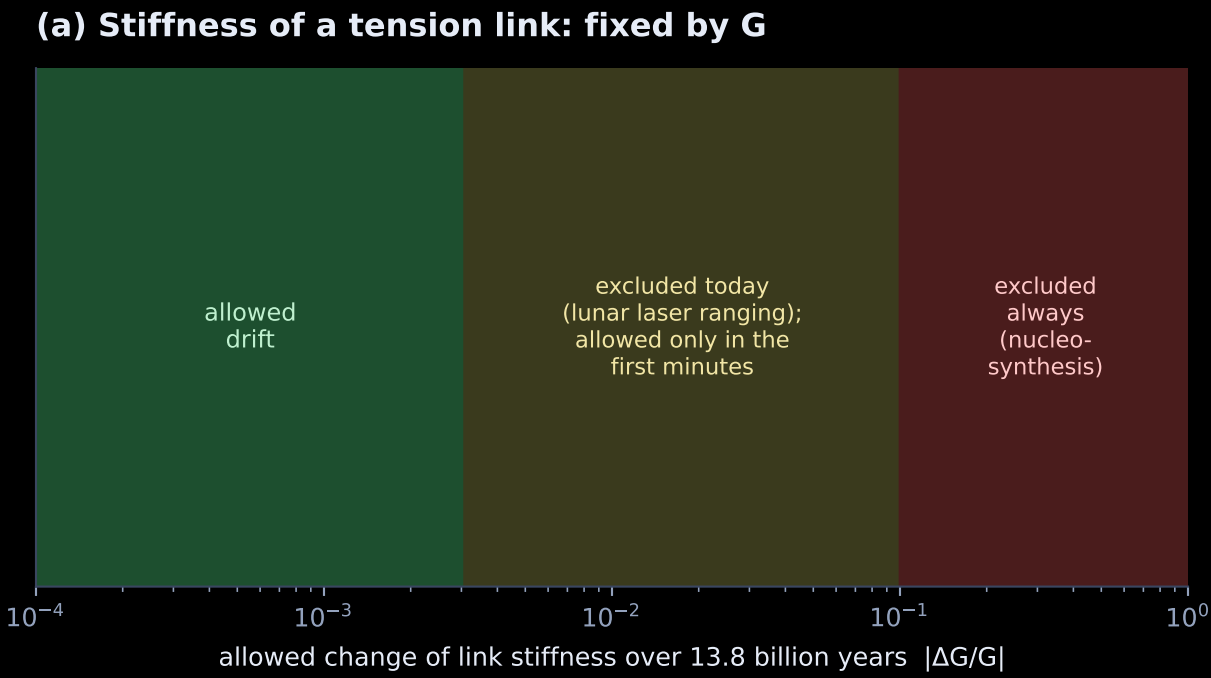


Figure 12. The grid's elasticity and its limits: how stiff, how tense, how far it can be squeezed and stretched

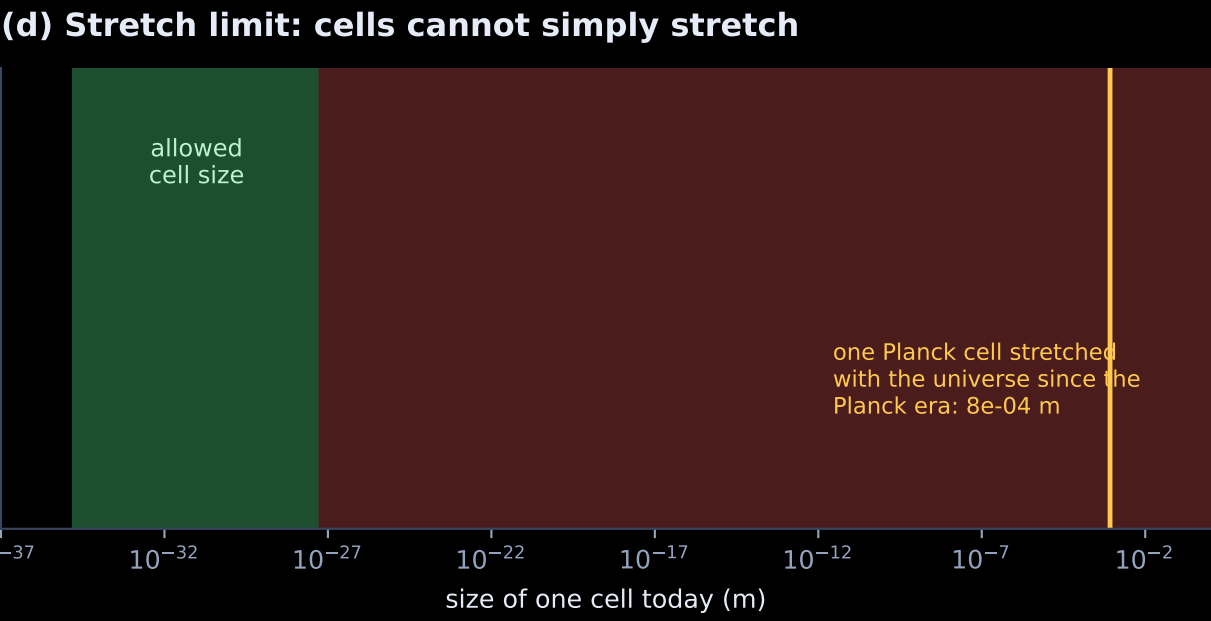
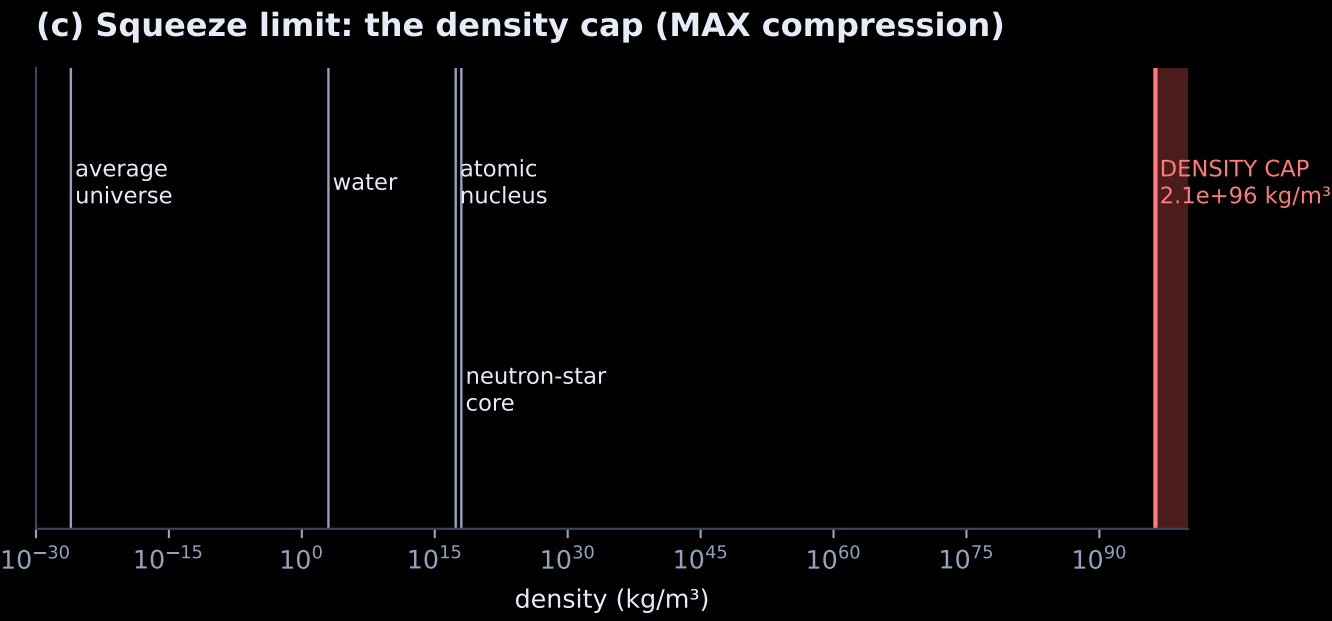


link tension scale $F = c^4/G = 1.21e + 44$ N (Planck force); stiffness of space $c^4/8\pi G = 4.8e + 42$ N

drift: $\dot{G}/G = (7.1 \pm 7.6) \times 10^{-14} \text{ yr}^{-1} \Rightarrow |\Delta G/G| < 3.1e - 03$ (2σ) over the age of the universe

today: $\rho_{DEC^2} = 5.3e - 10 \text{ J/m}^3$ (= tension per area per length, Pa);

law: $d \ln \rho_{DE} / d \ln a = \frac{1}{2} q$, $q = -\ddot{a} a / \dot{a}^2$; MAX where $q = 0$ (the Turnover Point)



$\rho_{cap} \approx 0.41 \rho_P$, $\rho_P = c^5/\hbar G^2$; collapse obeys $H^2 = \frac{8\pi G}{3} \rho (1 - \rho/\rho_{cap})$:

at the cap the squeeze turns into a rebound (Bounce, Planck star). Assumed from loop quantum cosmology, not measured.

stretch since the Planck era = $T_P/T_{CMB} = 5.2e + 31$; allowed stretch per cell $\leq \ell_{max}/\ell_P = 3.5e + 07$

$\Rightarrow$ the grid must ADD cells as space expands (OPEN requirement of the model).